# MID TERM EXAMINATION

Paper: Mid Term Examination - Set A | Duration: 3 Hours | Max Marks: 52 | Date: 14-03-2026

*Total Questions: 15 | Answer all questions.*

**Name:** ____________________________ **Roll No:** ________________________

**Student Sign:** ___________________ **Teacher Sign:** ___________________

## SECTION A: MCQS

**1.** Which of the following is typically used to measure the efficiency of an algorithm in terms of its execution time as the input size grows? **[2 Marks]**

a) Space Complexity
b) Time Complexity
c) Data Complexity
d) Memory Complexity

**2.** Which principle does a Stack data structure follow? **[2 Marks]**

a) First-In, First-Out (FIFO)
b) Last-In, First-Out (LIFO)
c) Last-In, Last-Out (LILO)
d) First-In, Last-Out (FILO)

**3.** What does the acronym LIFO stand for, and which common data structure adheres to this principle? **[2 Marks]**

## SECTION A: SHORT ANSWERS

**4.** What is a Data Structure? **[2 Marks]**

**5.** What is the purpose of a pointer in programming? **[2 Marks]**

**6.** Give an example of a non-linear data structure. **[2 Marks]**

## SECTION A: MCQS

**7.** In a singly linked list, which operation has a time complexity of O(1) if the head pointer is known? **[3 Marks]**

a) Searching for an element
b) Deleting the last element
c) Inserting an element at the beginning
d) Deleting an element from the middle

**8.** What is a primary advantage of a linked list over an array for dynamic data storage? **[3 Marks]**

a) Direct access to any element by index

b) Better cache performance due to contiguous memory allocation
c) Dynamic size, allowing easy insertion and deletion without reallocation overhead
d) Requires less memory per element

**9.** The postfix expression for the infix expression A * (B + C) / D is: **[3 Marks]**

a) ABC+*D/
b) AB*C+D/
c) A+B*C/D
d) A*B+C/D

## SECTION A: SHORT ANSWERS

**10.** State one significant advantage of a Doubly Linked List over a Singly Linked List. **[3 Marks]**

## SECTION A: MCQS

**11.** Explain the Big-O, Big-Omega, and Big-Theta notations used in algorithm analysis. **[5 Marks]**

## SECTION A: SHORT ANSWERS

**12.** Differentiate between an array and a linked list based on memory allocation and element access. **[5 Marks]**

## SECTION A: MCQS

**13.** What are the primary design considerations when implementing a circular queue using an array? How do these considerations address potential issues? **[6 Marks]**

## SECTION A: SHORT ANSWERS

**14.** Outline the steps to insert a new node with data X at the k-th position (1-indexed) in a singly linked list, assuming the head of the list is head. **[6 Marks]**

**15.** Convert the infix expression (A + B) * C - D / E to its equivalent postfix expression, showing the stack and output at each significant step. **[6 Marks]**

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**--- End of Question Paper ---**